StreamPlus Premium-Tier Upgrade Targeting

Predictive Modeling for Revenue Growth and Churn-Aware Campaign Targeting

Executive Summary

StreamPlus is a fictional direct-to-consumer streaming service seeking to increase average revenue per user (ARPU) by converting Basic subscribers to Premium. The upgrade creates an estimated $96 in incremental annual revenue, but irrelevant offers create churn risk: a non-converting recipient is assigned an expected $55 lifetime-value (LTV) cost. Historical untargeted campaigns convert only about 11% of recipients. The project therefore treats campaign targeting as a business-value optimization problem rather than a simple classification exercise.

A logistic regression model was trained on subscriber engagement and behavioral features. The model was evaluated on a stratified 30% holdout set and then converted into an operational targeting rule by testing probability thresholds against the business cost function Net Value = (True Positives x $96) - (False Positives x $55). The best holdout-set threshold was 0.31, generating an estimated $5,147 in Net LTV value and identifying substantially more true upgraders than the default 0.50 cutoff.

Headline recommendation

Target Basic subscribers only when predicted Premium-upgrade probability is at least 31%, then validate the policy through a controlled A/B test before production rollout.

1. Business Scenario and Decision Problem

StreamPlus serves roughly 4.9 million paying subscribers in North America. Basic is priced at $9 per month and Premium at $17 per month, making each successful upgrade worth approximately $96 of incremental annual subscription revenue. Broad campaigns, however, can annoy subscribers who perceive the message as irrelevant. Internal retention assumptions in the case estimate a 2.5 percentage-point increase in 30-day cancellation risk for a non-accepting recipient, which is translated into a $55 expected LTV loss per false-positive target.

- True Positive: subscriber is targeted and upgrades -> +$96 incremental annual revenue.
- False Positive: subscriber is targeted but does not upgrade -> -$55 expected LTV impact.
- False Negative: likely upgrader is not targeted -> missed revenue opportunity, but no added churn exposure.
- True Negative: subscriber is correctly left untargeted -> no incremental value or targeting cost.

This asymmetry means a model that maximizes accuracy is not necessarily the best campaign model. The useful question is: at what predicted probability does the expected value of contacting a subscriber become attractive enough to justify the churn risk?

2. Data Preparation and Quality Review

The dataset contains 2,500 subscriber records and eight numerical fields. All 2,500 rows are complete, so no missing-value imputation was required. Subscriber ID is retained only as an identifier and excluded from modeling. The target variable, accepted_upgrade, is moderately imbalanced: 777 subscribers accepted the upgrade (31.08%) and 1,723 declined (68.92%).

Measure	Result
Rows	2,500
Columns	8
Missing values	0
Accepted upgrades	777 (31.08%)
Declined upgrades	1,723 (68.92%)
Model features	6 behavioral / engagement variables
Train / test split	70% / 30% stratified

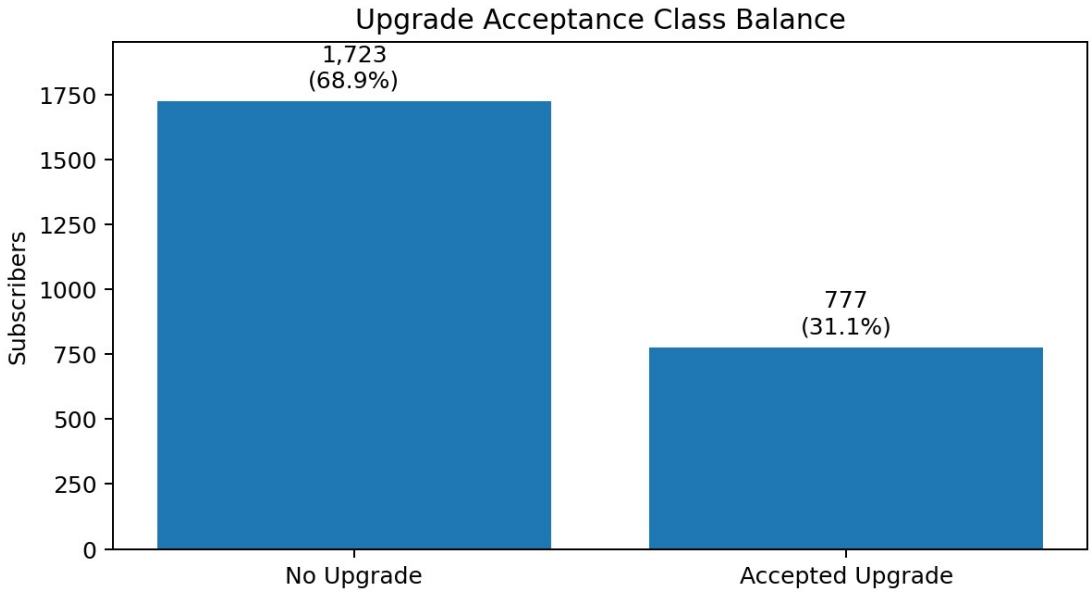


Figure 1. Target-class balance in the 2,500-subscriber dataset.

3. Model-Ready Data and Descriptive Statistics

A sample of the cleaned/model-ready table is shown below. Subscriber ID is omitted because it has no predictive meaning.

Months	Hours/wk	Devices	Search rate	Past offers	Age code	Accepted
29	8.2	1	0.196	2	2	0
29	5.6	3	0.317	3	4	1
16	11.7	1	0.486	0	2	1
5	5.0	4	0.259	1	3	0
6	6.1	1	0.046	1	3	0

Table 1. Model-ready sample (first five records).

Feature	Mean	Std. dev.	Median	Min	Max
Months subscribed	14.16	13.89	10.00	1	72
Content hours/week	8.96	5.17	8.00	0.60	44.50
Devices used	2.23	1.11	2.00	1	5
Recent search rate	0.197	0.146	0.166	0.000	0.845
Past offer responses	0.80	0.90	1.00	0	6
Age range code	2.43	1.02	2.00	1	4

Table 2. Selected descriptive statistics from the cleaned subscriber dataset.

4. Exploratory Findings

Accepted upgraders show a more engaged behavioral profile. On average, accepted subscribers had been subscribed for 19.47 months versus 13.01 months for decliners, watched 8.51 versus 5.93 hours per week, and used 2.39 versus 1.92 devices. These are descriptive associations rather than causal effects, but they are directionally consistent with the business hypothesis that more engaged and established customers are more receptive to a Premium offer.

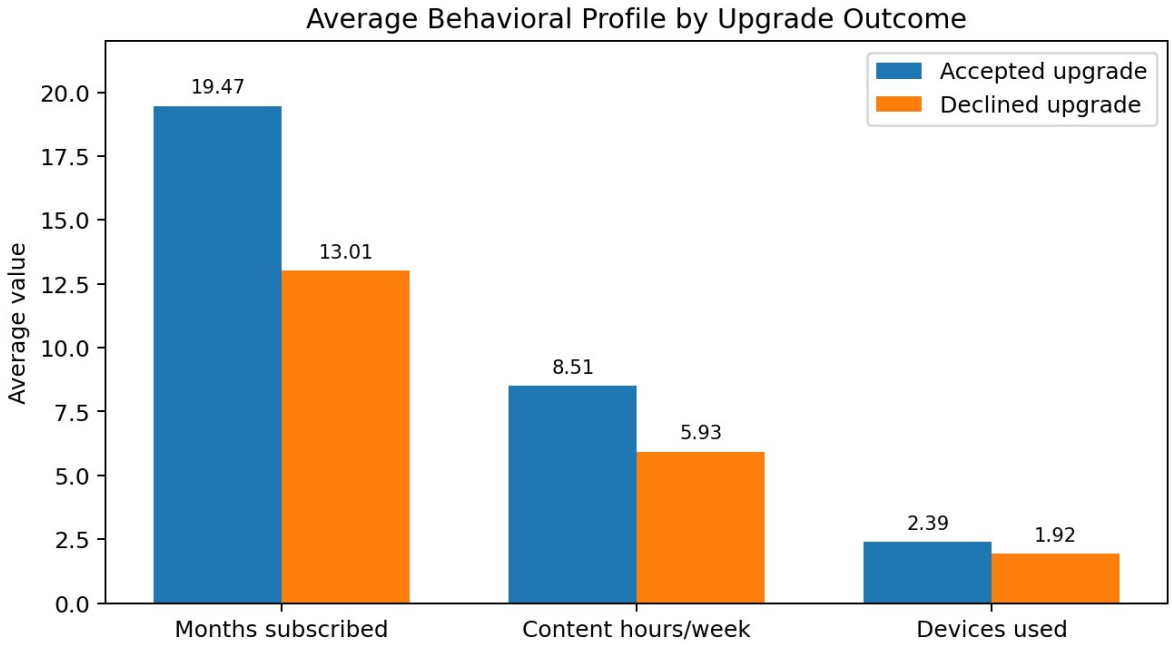


Figure 2. Average behavioral profile by upgrade outcome, regenerated for the cleaned portfolio analysis.

5. Predictive Model and Interpretation

A logistic regression classifier was selected because the campaign requires interpretable probabilities rather than only hard class labels. The six predictive features were months_subscribed, content_hours_per_week, devices_used, recent_search_rate, past_offer_responses, and age_range_code. The model was trained on 1,750 subscribers and evaluated on a stratified holdout set of 750 subscribers.

Exponentiating each logistic coefficient produces an odds ratio. Values above 1 indicate higher odds of upgrade as the feature increases; values below 1 indicate lower odds. These effects are per one-unit change and should not be compared as if all features share the same scale.

Feature	Coefficient	Odds ratio	Direction
recent_search_rate	2.803	16.501	Higher odds
past_offer_responses	0.492	1.635	Higher odds
devices_used	0.170	1.185	Higher odds
content_hours_per_week	0.031	1.031	Higher odds
months_subscribed	0.007	1.007	Higher odds
age_range_code	-0.174	0.840	Lower odds

Table 3. Logistic-regression coefficients and odds ratios.

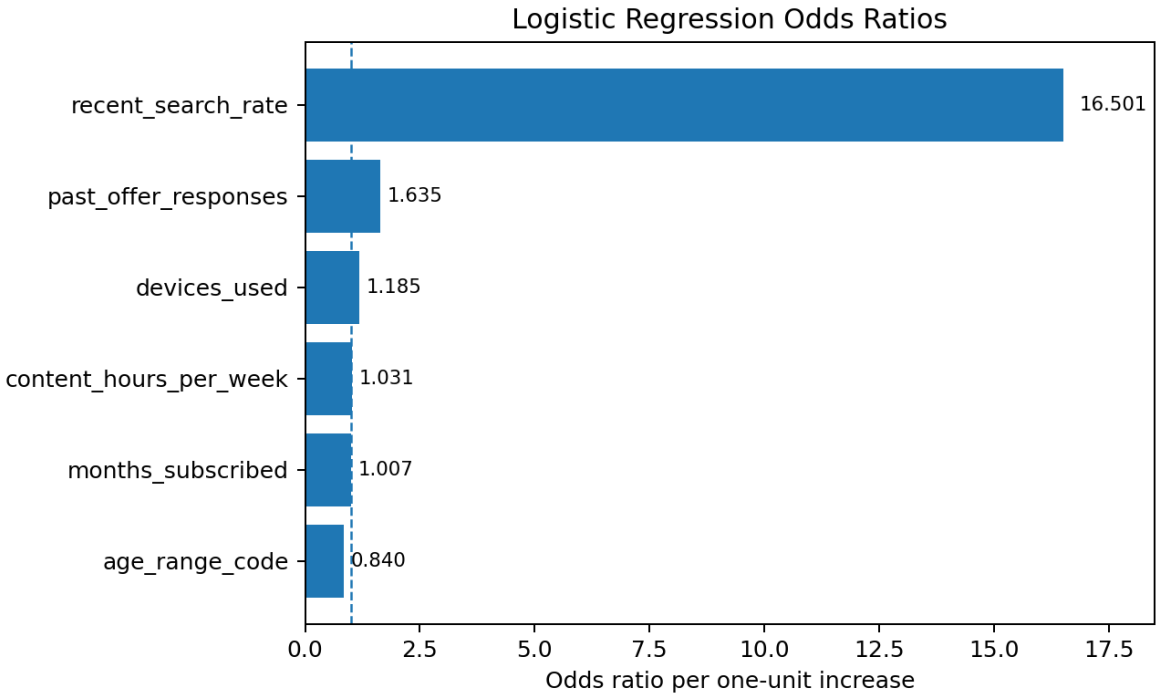


Figure 3. Odds ratios from the logistic regression model.

6. Baseline Classification Performance

Using the default 0.50 probability cutoff, the model achieved 72.0% accuracy and 62.9% precision, but recall was only 24.0%. The confusion matrix shows why: the model correctly identified 56 upgraders but missed 177. It produced only 33 false positives, so the cutoff is conservative. That may look attractive from a precision standpoint, yet it leaves a large number of profitable upgrade opportunities untouched.

Metric	0.50 Threshold
Accuracy	72.00%
Precision	62.92%
Recall	24.03%
F1 score	34.78%
True negatives	484
False positives	33
False negatives	177
True positives	56
Estimated Net Value	$3,561

Table 4. Baseline holdout performance at the default 0.50 classification cutoff.

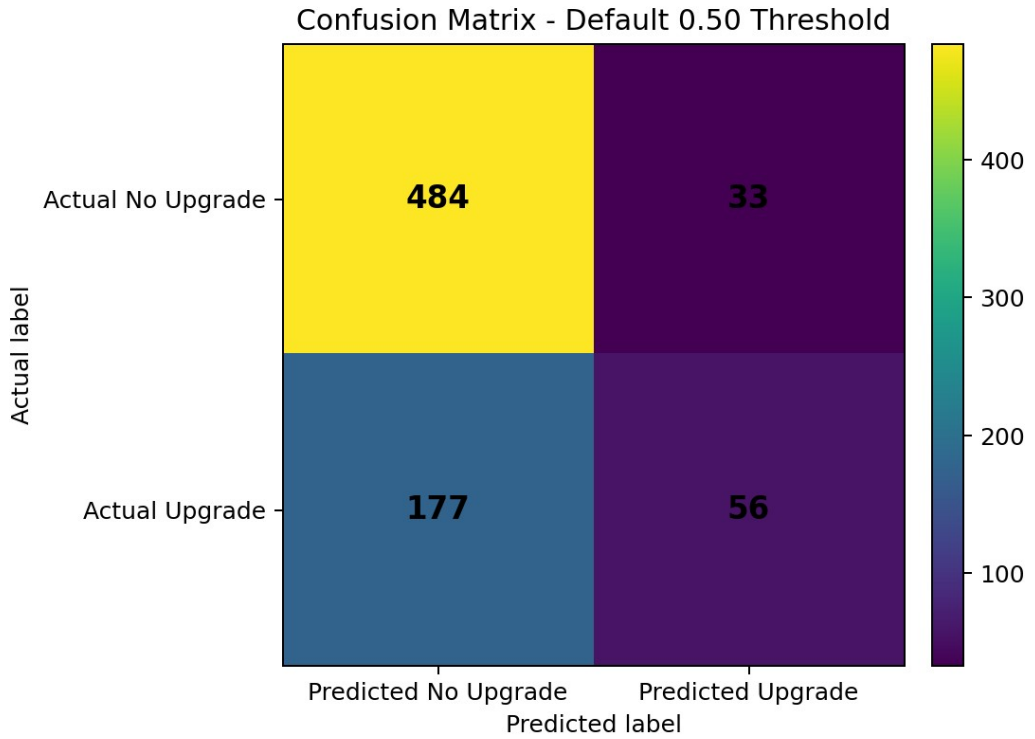


Figure 4. Holdout-set confusion matrix at the default 0.50 threshold.

7. Threshold Optimization Using Business Value

Because the false-positive cost and true-positive benefit are known, the model is more useful when its probabilities are converted into an economic targeting rule. The threshold was varied and Net Value was recalculated on the holdout set. The optimal tested threshold was 0.31, where 147 true positives and 163 false positives generated $5,147 of estimated Net LTV value.

At 0.31, precision falls to 47.42% but recall rises to 63.09%. This is the critical business insight: maximizing precision alone would be too conservative. The additional true positives captured at 0.31 more than compensate for the additional false-positive cost.

Threshold	TP	FP	FN	Precision	Recall	Net Value
0.30	151	176	82	46.18%	64.81%	$4,816
0.31	147	163	86	47.42%	63.09%	$5,147
0.32	138	150	95	47.92%	59.23%	$4,998
0.39	97	84	136	53.59%	41.63%	$4,692
0.45	75	48	158	60.98%	32.19%	$4,560
0.50	56	33	177	62.92%	24.03%	$3,561

Table 5. Selected holdout thresholds illustrating the precision-recall-value trade-off.

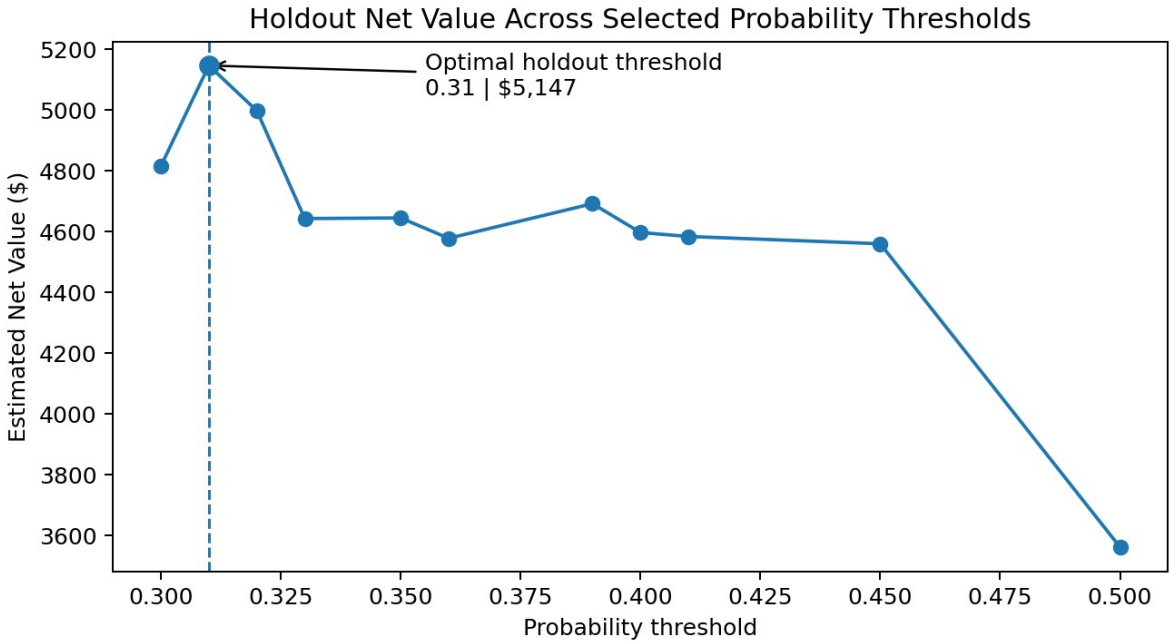


Figure 5. Holdout Net Value across selected probability thresholds; 0.31 is the best observed decision rule.

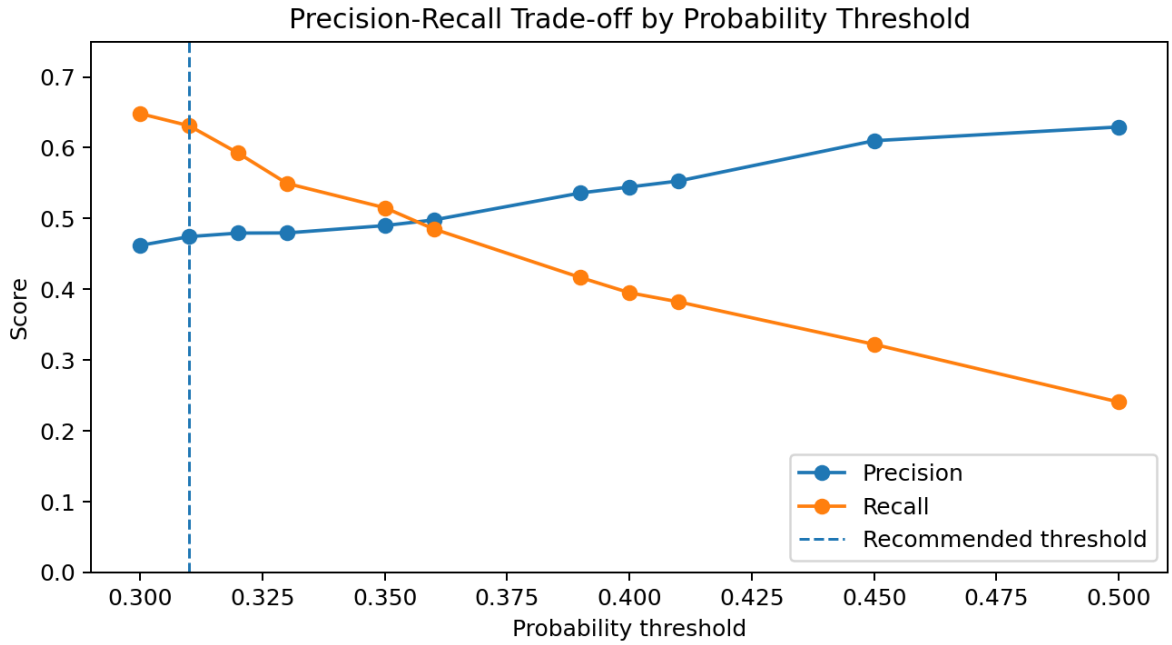


Figure 6. Precision-recall trade-off as the targeting threshold changes.

8. Recommended Targeting Rule and Historical Simulation

Operational rule

Send the Premium offer only when a Basic subscriber has a predicted upgrade probability of at least 0.31. The 0.31 cutoff is selected from held-out test performance, not from the full historical dataset.

On the 750-subscriber holdout set, this rule would contact 310 people and yield an estimated 47.42% conversion rate among contacted subscribers, 63.09% recall, and $5,147 of Net LTV impact. Applying the same rule retrospectively to all 2,500 historical subscribers would contact 1,060 people, produce an in-sample conversion rate of 46.23%, recall of 63.06%, and approximately $15,690 of Net LTV impact.

Measure	Holdout @ 0.31	Full historical @ 0.31
Subscribers evaluated	750	2,500
Subscribers contacted	310	1,060
Conversion rate / precision	47.42%	46.23%
Recall	63.09%	63.06%
Estimated Net LTV impact	$5,147	$15,690
Interpretation	Out-of-sample evidence	In-sample sizing simulation

Table 6. Recommended holdout result versus full-history simulation.

A useful caution emerges from the full-data simulation: a 0.39 threshold produces approximately $16,333 of in-sample Net Value, higher than 0.31. That does not justify changing the recommendation. Because the model was trained on most of the same records, optimizing directly on the full sample would introduce optimistic bias. The 0.31 rule is preferable because it was chosen using held-out data and is therefore more defensible as a future-policy estimate.

9. Recommendations

- Replace broad, untargeted upgrade campaigns with model-scored targeting using the 0.31 probability threshold.
- Use Net LTV Impact as the primary campaign decision metric; precision, recall, and F1 should remain supporting diagnostics.
- Monitor false positives carefully because the business case assumes they create measurable churn-related LTV damage.
- Run a randomized A/B test comparing the model-targeted policy against the existing untargeted approach before a full production rollout.
- Recalculate the optimal threshold when pricing, upgrade economics, churn assumptions, or subscriber behavior changes.
- Track calibration and model drift; probabilities must remain reliable if they are used directly for campaign decisions.

10. Limitations and Future Work

- The scenario is fictional and the $96 benefit, $55 false-positive cost, and churn assumptions are case inputs rather than independently validated causal estimates.
- The model predicts upgrade acceptance but does not estimate the causal lift created by sending an offer. High-propensity customers may have upgraded without intervention.
- The feature set is limited to six behavioral variables and excludes detailed content preferences, payment history, support interactions, and richer customer demographics.
- The accepted-upgrade class is the minority class, so accuracy can obscure practical performance; threshold-aware metrics are more informative.
- Only logistic regression is used in the core model. Tree-based models, gradient boosting, and calibrated probability models could capture nonlinear patterns.
- A stronger next step would be uplift modeling or treatment-effect estimation, followed by controlled experimentation and real-time monitoring.

11. Conclusion

The project shows how a classification model can be translated into a business policy. The default 0.50 cutoff offers high precision but misses too many potential upgraders. When the decision threshold is optimized against explicit campaign economics, 0.31 produces the strongest holdout-set Net Value. The resulting rule is simple and actionable: score Basic subscribers, contact those above the threshold, and continuously validate whether realized conversion and churn outcomes match the assumed economics.

The core takeaway is that the best model decision is not necessarily the one with the highest accuracy or precision; it is the one that creates the most defensible expected business value.

Source note: Figures and tables in this report were regenerated from the cleaned StreamPlus portfolio notebook and the underlying project outputs.